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SESSION — Oct 29 & 30

10 · 60 · 10

Agents, Subagents, and Cost



GOAL —

Participants understand what an agentic workflow is, when splitting work across subagents helps, and why parallelism buys speed rather than judgment. Before typing, they ask three questions: What is this task worth? Which model or effort setting fits it? How much verification will the output demand?

WHY —

This is where "chatbot" becomes "tool," and also where errors multiply. Nate's point: firing off ten agents makes the work faster and harder to check, and no better. The same judgment applies one level down. Students default to one model at one setting for everything; fast cheap models are right for half of research work, and slow expensive ones are wrong for half of it too. "Cost" means money, wall-clock time, and how much checking the output will need.

READY-MADE MATERIALS —

Use these when nobody has their own. Everything here is public or was written for this course.

The three documents

The last three annual editions of the Department of Defense report to Congress on Military and Security Developments Involving the People's Republic of China (public, on defense.gov). Same topic, same structure, changing numbers.

The agentic task

Read these three reports. Extract every quantitative claim about the size of the PLA Navy into a table: claim, report year, page cite. Flag every place the editions disagree with each other. The implicit decision to catch: what counts as a ship.

Manual version for tools without agents

Assign one report per person, same table, then merge. The merge is where the parallel-reads lesson lands.

Question to run at two settings

A mid-sized city is considering a 1% payroll tax to fund a homelessness program. What are the three strongest objections an economist would raise, and what evidence would settle each? Cheapest model, then highest effort. Time both.

Five tasks for the cost estimation game, with the tier to argue toward

(1) Turn this bulleted list into a table: A. (2) Find the five most-cited papers on secondary sanctions since 2018: B, since you will verify every one. (3) Draft the limitations section of a survey-based study: C. (4) Explain fixed effects versus random effects in two sentences: A, or B if you are going to teach it. (5) Read these three DoD reports and tell me where they disagree: B and agentic; the disagreements need checking. Expect a fight over 4 and 5.

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MIN

COLD OPEN

Data homework demo.

60

MIN

CORE

1 What an agent is 10

Screen-share a simple diagram: an agent is a model in a loop with tools. It reads, acts, observes, repeats. Every step is a chance to be wrong; every step is also something you can inspect.

2 One task, agentially 15

Everyone gives the model the three-report extraction task with file access. Watch it work. Note where it skipped. Show how to read the trace, step by step; if your tool hides it, that's a limitation worth knowing.

3 The intern analogy 10

When would you split a research task across five interns? (Independent reads: five countries, five years.) When would that make it worse? (Anything requiring a shared understanding of the argument.) Subagents are the same. Reading parallelizes; writing and judgment don't. The real win connects to Session 2: each subagent gets a fresh context, so the main thread doesn't rot, and that only holds if the subtasks are truly independent.

4 The three tiers 15

Build a task ladder live in the shared doc. **Tier A, disposable:** reformatting, quick definitions, "what's another word for." Cheapest, fastest model, minimal checking. **Tier B, consequential but checkable:** lit search scoping, first-draft summaries, code you'll run and inspect. Mid-tier model or extended thinking; verify the parts you'll use. **Tier C, judgment-heavy:** methods critique, framing a research question, anything going into a deliverable under your name. Best model, high effort, and the output is a draft you own, not an answer. Then everyone runs one real question at the cheapest setting and the highest, and times both. Note the differences honestly; sometimes there aren't any.

5 Cost estimation game 10

Read out five tasks; everyone types a tier (A, B, or C) into chat at the same time. Argue the borderline ones. Where does "agentic" sit on the ladder? Usually B, sometimes C, never A.

10

MIN

DEBRIEF

What did the agent skip? Where did it decide something you'd have wanted to decide? Where did the expensive setting earn its cost, and where was it identical? Log both. Start a "tiering guide" section in the Playbook.

HOMEWORK

Run one agentic task and one manual version of the same task. Which was faster? Which was right? Tag each AI task you do this week A/B/C before you run it, and notice when you were wrong.

60

MIN CUT

FOR 60 MINUTES

Cut the cost game; fold the tiers into a five-minute framing statement after the intern analogy.

FACILITATOR WATCH-OUTS

- The agent will make an implicit decision (which claims count as "quantitative," how to resolve a conflict). Catch one live; that's the lesson.
 - If participants don't have agentic tools, simulate: run the steps manually in sequence and discuss where a loop would have helped and where it would have hidden a mistake.
 - The interesting finding on tiers is usually that the cheap model was fine for more than expected. Let that land; don't rush to defend the frontier model.
 - If RAND's tool exposes only one model, run the tier exercise on effort/thinking settings and conversation length instead.
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